Can Cryptocurrencies Help Predict Volatility Movements in Financial Markets?

Bayesian Stochastic Volatility Model via Hamiltonian Monte Carlo

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CHAPTER ONE: The data

The dataset consists of 122 monthly closing observations spanning from April 2016 to May 2026, and includes four financial instruments: - VIX; - BTC; - ETH; - GLD.

The objective is to determine whether past movements in BTC, ETH, and GLD can successfully predict market volatility, as measured by the VIX.

1.1 Descriptive statistics

```
#descriptive statistics

##   Financial_instrument        Mean      Std_dev   Min     Max
## 1                  BTC 2019726.693 2243773.6580 44850 9640570
## 2                  ETH  107187.519  123779.1139   800  462890
## 3                  GLD   18424.902    8039.2697 10961   48375
## 4                  VIX    1852.303     691.9118   951    5354
```

1.2 Transformation in log-returns

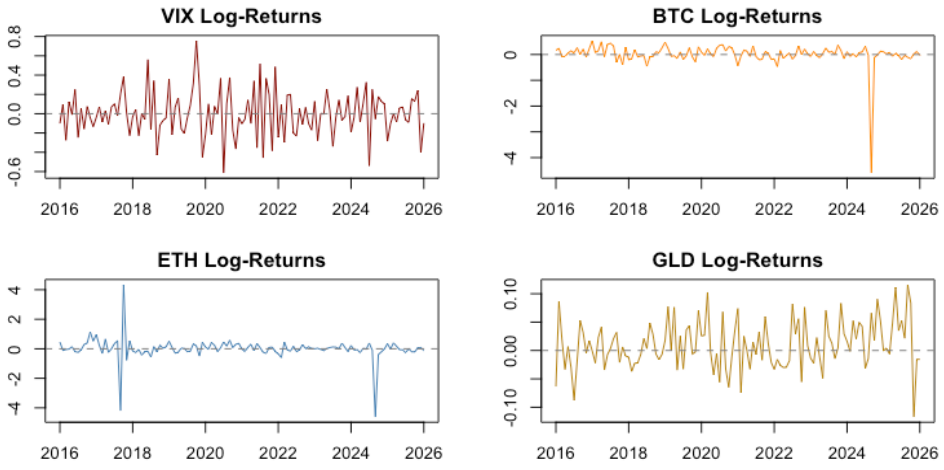
$$y_t = \log(\text{VIX}_t) - \log(\text{VIX}_{t-1})$$

For the covariates (BTC, ETH, GLD) we use log-returns:

$$r_t = \log(P_t) - \log(P_{t-1})$$

The covariates are lagged by 12 months: the predictor at time t is the log-return observed 12 months earlier r_{t-12}. This tests whether past movements in crypto and gold can anticipate future VIX volatility with a one-year horizon.

1.3 Time series graphs



CHAPTER 2: Stochastic Volatility Model with covariates

The Stochastic Volatility (SV) model with covariates consists of two equations:

Measurement equation (or Observation equation):

$$y_t = \exp(h_t/2)\,\varepsilon_t, \qquad \varepsilon_t \overset{iid}{\sim} \mathcal{N}(0,1)$$

where $y_t = \log(\text{VIX}_t) - \log(\text{VIX}_{t-1})$ are the log-returns of the VIX. The instantaneous volatility is $\exp(h_t/2)$, where h_t is the latent log-volatility at time t: it is not directly observable and is estimated by the MCMC algorithm.

State equation (or Log-volatility dynamics equation):

$$h_t = \mu + \phi(h_{t-1} - \mu) + \beta_1 x_{1,t-12} + \beta_2 x_{2,t-12} + \beta_3 x_{3,t-12} + \sigma_\eta \eta_t, \qquad \eta_t \overset{iid}{\sim} \mathcal{N}(0,1)$$

This equation says that the log-volatility today (h_t) depends on:

- μ: the long-run mean level of log-volatility;
- $\phi(h_{t-1} - \mu)$: the persistence component;
- $\beta_1 x_{1,t-12} + \beta_2 x_{2,t-12} + \beta_3 x_{3,t-12}$: the effect of lagged covariates (BTC, ETH, GLD at $t - 12$);
- $\sigma_\eta \eta_t$: unpredictable stochastic shock.

The initial condition is:

$$h_1 \sim \mathcal{N}\left(\mu, \frac{\sigma_\eta^2}{1 - \phi^2}\right)$$

which corresponds to the stationary distribution of the AR(1) process.

2.1 Parameters to estimate

The vector of structural parameters is:

$$\boldsymbol{\theta} = (\mu,\ \phi,\ \sigma_\eta,\ \beta_1,\ \beta_2,\ \beta_3)$$

In addition to $\boldsymbol{\theta}$, we also estimate the vector of latent states $\mathbf{h} = (h_1, \dots, h_T)$ of dimension $T = 109$.

2.2 Prior Distributions

We choose priors that respect the constraints of the model:

$$\mu \sim \mathcal{N}(0, 10^2) \qquad \text{(weakly informative prior)}$$

$$\phi \sim \text{Uniform}(-1,1) \qquad \text{(stationarity constraint)}$$

$$\sigma_\eta \sim \text{HalfNormal}(0,1) \qquad \text{(positivity contraint)}$$

$$\beta_1, \beta_2, \beta_3 \sim \mathcal{N}(0,1) \qquad \text{(weakly informative priors)}$$

CHAPTER THREE: The Algorithm: HMC/NUTS via Stan

3.1 From MH to Hamiltonian Monte Carlo

The Block Metropolis-Hastings algorithm explored the posterior by proposing random Gaussian jumps and accepting or rejecting them. Its main limitation is that the proposal scale δ_j must be tuned manually, and for correlated parameters like ϕ and σ_η the chain mixes slowly.

Hamiltonian Monte Carlo (HMC) solves this by exploiting the gradient of the log-posterior. Instead of a random walk, HMC simulates a physical system where the parameter vector $\boldsymbol{\theta}$ is a particle moving under the influence of a potential energy $U(\boldsymbol{\theta}) = -\log\pi(\boldsymbol{\theta} \mid \mathbf{y})$.

At each step:

1. A random momentum vector $\mathbf{p}$ is drawn from $\mathcal{N}(\mathbf{0}, \mathbf{M})$, where M is a mass matrix which regulates the inertia and the scale of movements in the various directions;

2. The particle follows Hamilton's equations for L leapfrog steps of size ε:

$$\mathbf{p} \leftarrow \mathbf{p} - \frac{\varepsilon}{2}\nabla U(\boldsymbol{\theta}), \qquad \boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \varepsilon\mathbf{M}^{-1}\mathbf{p}$$

The position $\boldsymbol{\theta}$ is then updated by projecting the particle forward, leveraging the newly obtained velocity $\varepsilon\mathbf{M}^{-1}\mathbf{p}$.

3. The proposed $(\boldsymbol{\theta}^*, \mathbf{p}^*)$ is accepted with probability $\min\{1, \exp\big(-H(\boldsymbol{\theta}^*, \mathbf{p}^*) + H(\boldsymbol{\theta}, \mathbf{p})\big)\}$

The mathematics: $H(\boldsymbol{\theta}, \mathbf{p})$ represents the **Hamiltonian**, which is the total energy of the system (potential energy provided by the posterior distribution + kinetic energy provided by the momentum trigger).

If theoretical physics were perfectly simulated, the final energy H^* would be identical to the initial energy H, and the acceptance probability would be exactly $= 1(100\%)$.

Since the simulation introduces small numerical errors due to the step size ε, the total energy may fluctuate slightly.

This filter corrects the mathematical error: if the error is microscopic, the proposal is accepted; if an extreme energy deviation occurs (numerical instability), the proposal is rejected

The **No-U-Turn Sampler (NUTS)** is an adaptive extension that automatically selects the number of leapfrog steps L, stopping when the trajectory would start doubling back. This eliminates the need to tune L manually.

In practice, this means:

- No manual tuning of proposal scales;
- Much lower autocorrelation in the chains;
- Higher ESS for the same number of iterations;
- The warmup phase adapts the step size ε automatically.

3.2 Stan Model

The Stan model block encodes exactly the mathematical structure described before: the observation equation, the AR(1) latent volatility equation, and the priors.

```
#stan model
sv_stan_code <- "
data {
  int<lower=1> T;
  vector[T] y;
  vector[T] x1;
  vector[T] x2;
  vector[T] x3;
}

parameters {
  real mu;
  real<lower=-1, upper=1> phi;
  real<lower=0> sigma_eta;
  real beta1;
  real beta2;
  real beta3;
  vector[T] h_std;   // standardized latent states ~ N(0,1)
}

transformed parameters {
  vector[T] h;       // actual latent log-volatility
  // h_1: stationary distribution
  h[1] = mu + (sigma_eta / sqrt(1 - phi * phi)) * h_std[1];
  // h_t for t = 2..T: AR(1) with covariates
  for (t in 2:T) {
    h[t] = mu + phi * (h[t-1] - mu)
               + beta1 * x1[t] + beta2 * x2[t] + beta3 * x3[t]
               + sigma_eta * h_std[t];
  }
}

model {
  // Priors
  mu        ~ normal(0, 10);
  // phi: no explicit prior — Uniform(-1,1) implicit from <lower=-1,upper=1>
  sigma_eta ~ normal(0, 1);
  beta1     ~ normal(0, 1);
  beta2     ~ normal(0, 1);
  beta3     ~ normal(0, 1);

  h_std ~ std_normal();

  for (t in 1:T)
    y[t] ~ normal(0, exp(h[t] / 2));
```

```
}
"

#prepare data list for Stan
stan_data <- list(
  T  = T,
  y  = y,
  x1 = x1,
  x2 = x2,
  x3 = x3
)

#fit the model
fit <- stan(
  model_code = sv_stan_code,
  data       = stan_data,
  chains     = 4,
  iter       = 8000,
  warmup     = 4000,
  seed       = 42,
  refresh    = 1000,
  control    = list(
    adapt_delta   = 0.99,
    max_treedepth = 15
  )
)
```

CHAPTER FOUR: Results

4.1 Posterior Summary

```
##              mean se_mean     sd    2.5%     25%     50%     75%   97.5% n_eff
## mu        -3.0701  0.0043 0.3415 -3.7425 -3.2156 -3.0454 -2.8916 -2.5584  6355
## phi        0.5776  0.0050 0.3108 -0.2653  0.4394  0.6463  0.7999  0.9643  3860
## sigma_eta  0.4805  0.0032 0.2116  0.0713  0.3359  0.4754  0.6187  0.9178  4365
## beta1      0.4215  0.0033 0.3864 -0.3437  0.1687  0.4248  0.6762  1.1721 13536
## beta2     -0.3823  0.0026 0.2873 -0.9894 -0.5659 -0.3635 -0.1818  0.1275 12021
## beta3     -0.0842  0.0061 0.9764 -1.9804 -0.7546 -0.0907  0.5830  1.7984 25811
##              Rhat
## mu        1.0000
## phi       1.0004
## sigma_eta 1.0004
## beta1     1.0003
## beta2     1.0001
## beta3     0.9998
##
```

4.2 Potential Scale Reduction Factor diagnostics

When using Stan, the software does not run a single simulation but executes multiple independent simulations (chains). Each chain starts from a different, random initial point within the parameter space.

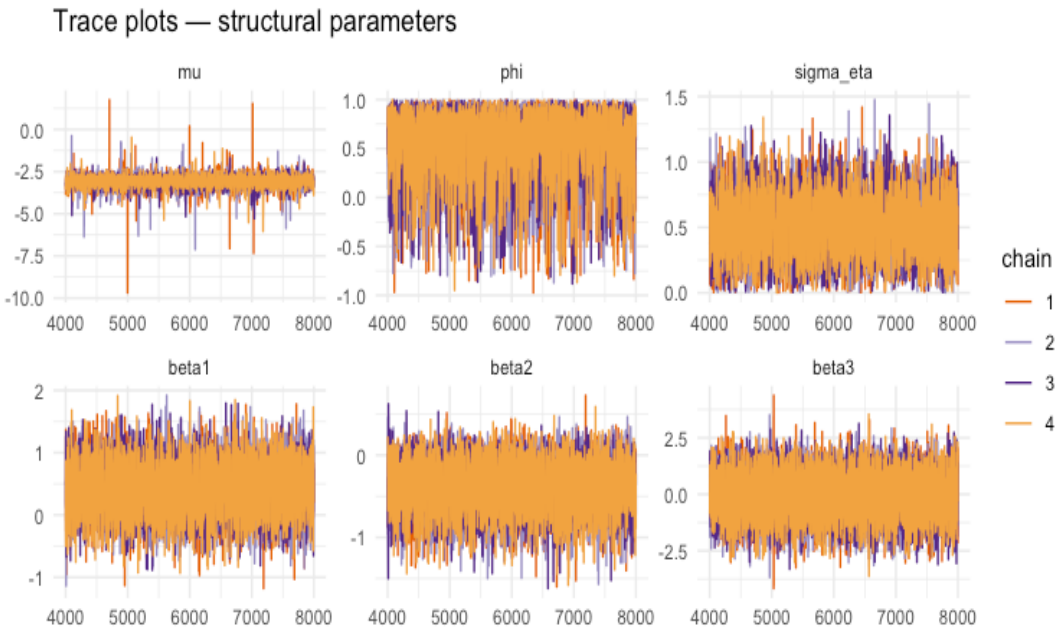
The $\hat{R}$ diagnostic compares two types of variability:

- Between-chain variance: how far apart the means of the 4 chains are from one another;
- Within-chain variance: how much each individual chain oscillates internally.

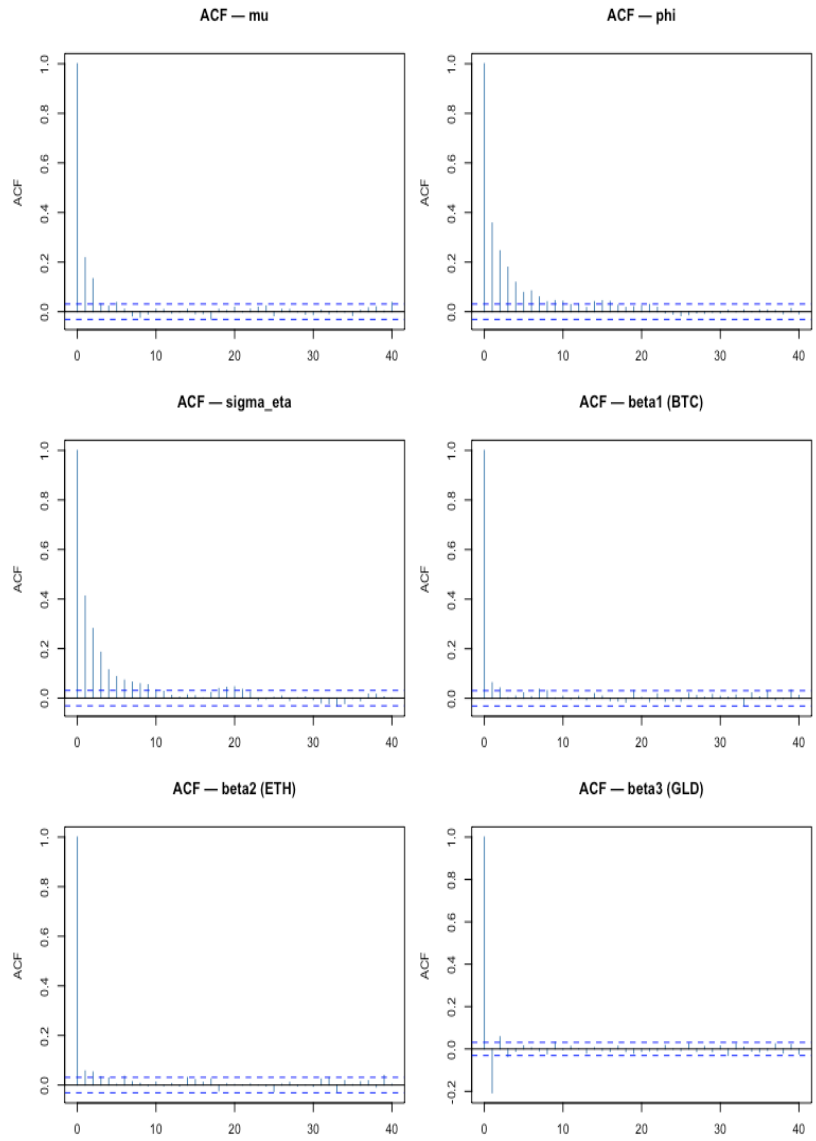
The $\hat{R}$ statistic is the ratio between these two variances. If this ratio is $= 1$, it means that the algorithm has reached a perfect convergence. If the value of the ratio is greater than 1.01 it means that the simulation is not stable and the chains do not find the correct distribution.

```
##           Parameter    Mean     SD Lower_bound Upper_bound   ESS   Rhat
## mu               mu -3.0701 0.3415     -3.7425     -2.5584  6355 1.0000
## phi             phi  0.5776 0.3108     -0.2653      0.9643  3860 1.0004
## sigma_eta sigma_eta  0.4805 0.2116      0.0713      0.9178  4365 1.0004
## beta1         beta1  0.4215 0.3864     -0.3437      1.1721 13536 1.0003
## beta2         beta2 -0.3823 0.2873     -0.9894      0.1275 12021 1.0001
## beta3         beta3 -0.0842 0.9764     -1.9804      1.7984 25811 0.9998
```

4.3 Trace Plots



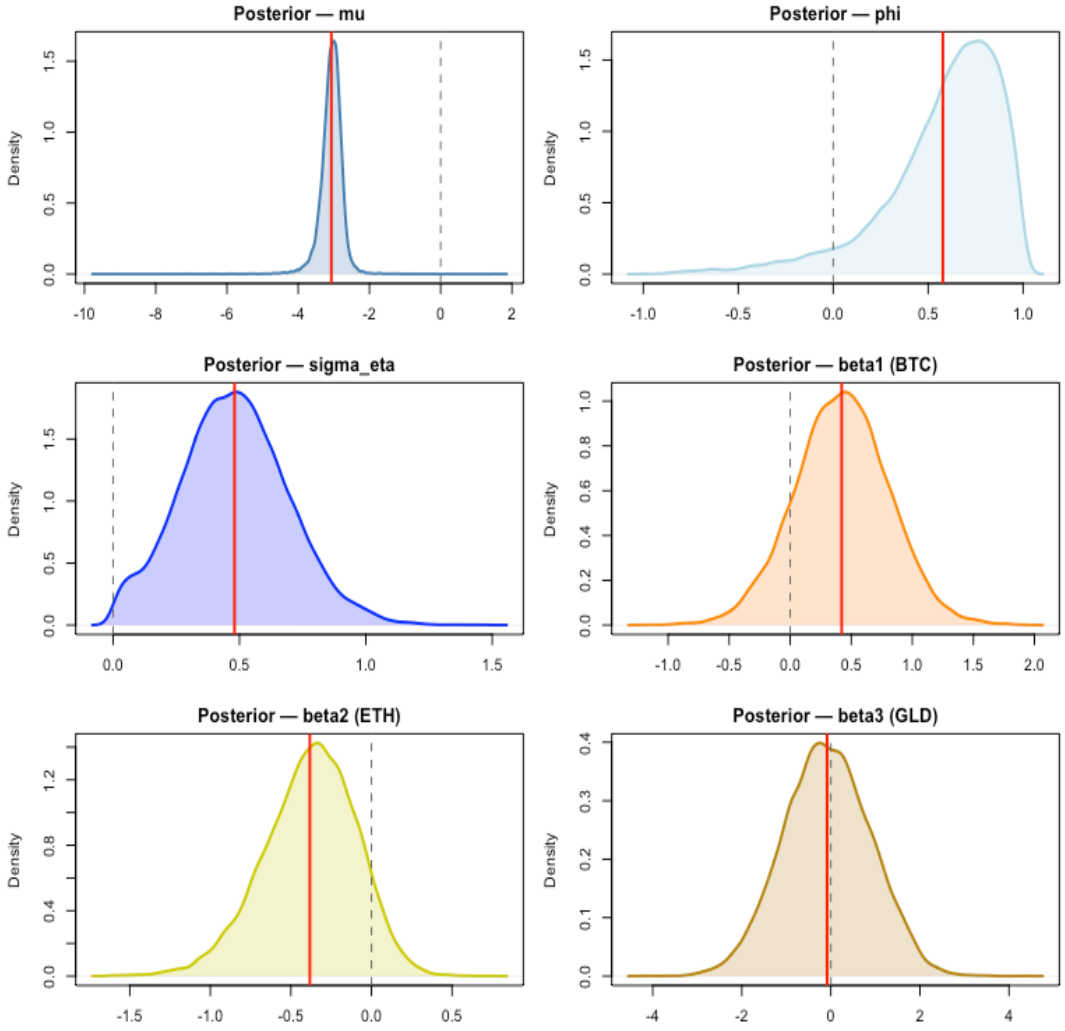
4.4 ACF Plots



4.5 Effective Sample Size (ESS)

```
##              ESS  Evidence
## mu          6355 Excellent
## phi         3860 Excellent
## sigma_eta   4365 Excellent
## beta1      13536 Excellent
## beta2      12021 Excellent
## beta3      25811 Excellent
```

4.6 Posterior Distributions

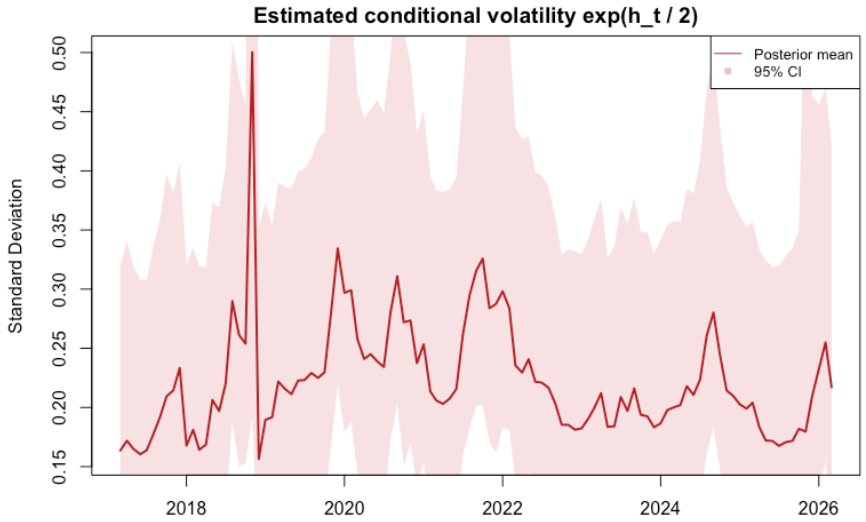


The VIX is primarily driven by its own dynamics (μ and ϕ), while BTC, ETH and GLD do not add significant predictive power.

4.7 Credible Intervals

```
##             Parameter    Mean  CI_2.5 CI_97.5              Excl_zero
## mu                 mu -3.0701 -3.7425 -2.5584                    Yes
## phi               phi  0.5776 -0.2653  0.9643                     No
## sigma_eta   sigma_eta  0.4805  0.0713  0.9178 Yes (constrained > 0)
## beta1     beta1 (BTC)  0.4215 -0.3437  1.1721                     No
## beta2     beta2 (ETH) -0.3823 -0.9894  0.1275                     No
## beta3     beta3 (GLD) -0.0842 -1.9804  1.7984                     No
```

4.8 Estimated Latent Volatility



At each point in time, $\exp(h_t/2)$ represents the estimated standard deviation of VIX returns conditional on the latent log-volatility h_t.

$\exp(h_t/2)$ is the standard deviation of y_t conditioned to h_t.

It comes from the model:

$$y_t = \exp(h_t/2)\,\varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0,1)$$

Consequently:

$$y_t \mid h_t \sim \mathcal{N}\big(0,\ \exp(h_t)\big)$$

Because:

$$\sigma^2 = [\exp(h_t/2)]^2 = \exp(h_t)$$

Several episodes stand out clearly. The most prominent peak occurs in early 2020, consistent with the extreme market dislocation caused by the Covid-19 pandemic.

A second elevated period is visible around 2021–2022, corresponding to post-pandemic inflationary pressures and monetary tightening. Interestingly, after a brief lull in 2023, the series registers a distinct volatility spike mid-2024, followed by an aggressive compression towards historical minima in 2025, before exhibiting a sharp upward rebound moving into early 2026.

The 95% credible band is notably wide throughout the entire sample, and widens further during stress episodes. This reflects genuine posterior uncertainty about h_t: when observed VIX returns are large and erratic, the model assigns more dispersion to the latent state.

The width of the band also reflects the limited sample size (109 monthly observations).

Conclusions

The Bayesian analysis of the Stochastic Volatility model with covariates, estimated via HMC/NUTS in Stan, produced the following main results.

- Convergence: all $\hat{R}$ values are below 1.01 (maximum: 1.0004), confirming that the 4 chains have converged to the same posterior distribution. The minimum ESS across structural parameters is 3860, well above the threshold of 400 recommended for reliable inference. Trace plot shows good mixing with no systematic trends.

- Process parameters: the estimated long-run mean of log-volatility is $\hat{\mu} =$ -3.07, corresponding to a baseline volatility of $\exp(\hat{\mu}/2) \approx 0.215$.

The persistence parameter is $\hat{\phi} = 0.578$: a positive value below 1 indicates that volatility mean-reverts toward its long-run level on a monthly horizon. The variation of volatility is $\hat{\sigma}_{\eta} = 0.481$.

- Effect of covariates: the estimated coefficients are $\hat{\beta}_1 = 0.421$ (BTC), $\hat{\beta}_2 =$ -0.382 (ETH), $\hat{\beta}_3 =$ -0.084 (GLD).

All 95% credible intervals include zero, indicating that none of the covariates has a statistically distinguishable effect on log-volatility dynamics at a monthly horizon with a 12-month lag.

This is an interpretable result: VIX volatility is primarily driven by its own AR dynamics, and past movements in crypto assets and gold do not add significant predictive power at this time scale.

- Latent volatility: the estimated volatility $\exp\left(\hat{h}_t/2\right)$ successfully captures known stress periods. The 95% credible band widens during periods of greater uncertainty, consistent with the stochastic nature of the model.